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Mechanical Engineering Project: First-Shot Acoustic Anomaly Detection Under Domain Shift

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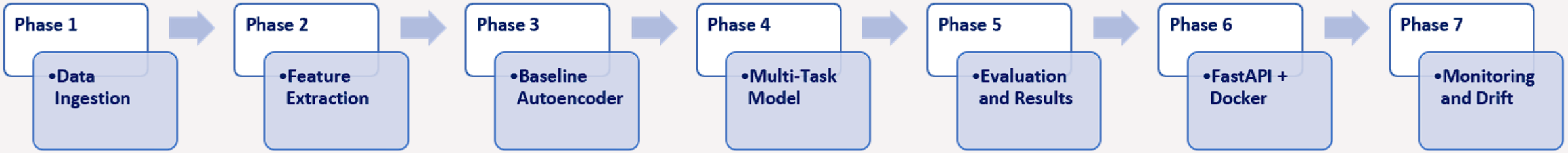
Project Overview

What We Built

A complete ML system that listens to 10-second industrial machine sounds and automatically determines whether the machine is normal or not normal .



7 Machine Types

ToyCar	ToyTrain	Bearing	Fan	Gearbox	Slider	Valve

The Challenge : Domain Shift

What is Anomaly Detection?

- Train ONLY on normal sounds (no anomaly examples needed)
- Model learns what 'normal' looks like for each machine
- At inference: high reconstruction error = something is wrong
- Detects NEW failure modes never seen during training

What is Domain Shift?

- Trained in one environment, tested in another
- Different microphone, placement, or recording setup
- Different background noise or operating conditions
- SOURCE domain: 990 training samples per machine
- TARGET domain: only 10 samples

Dataset Details

8400 WAV Files

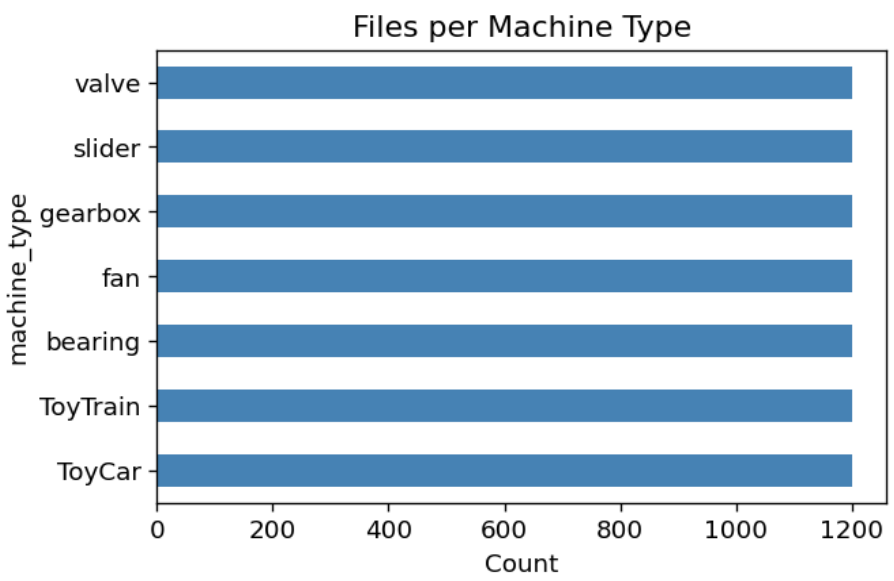
7 Machine Types

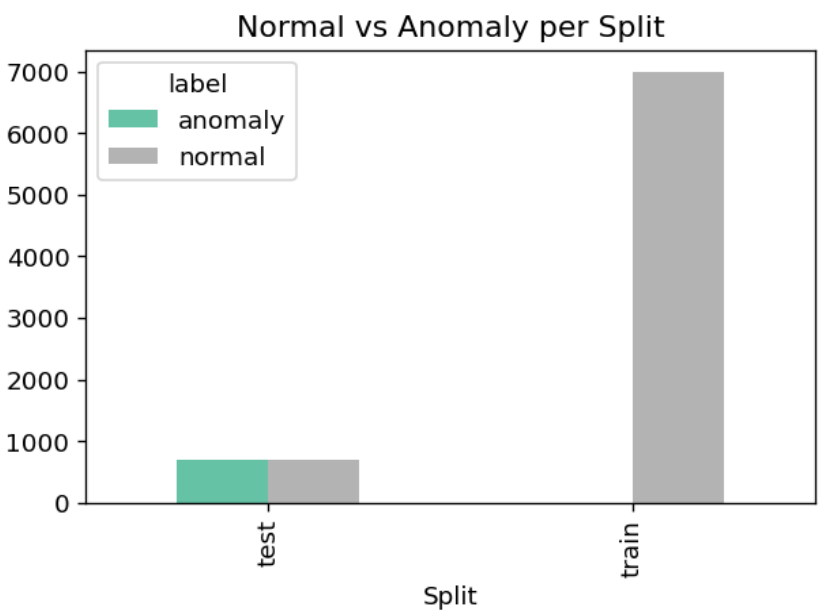
10 seconds / clip

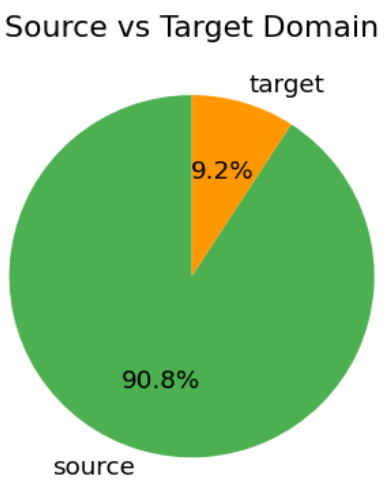
16 kHz Sample Rate

Source: Target Ratio
99 : 1

Dataset Distribution







Phase 1 : Data Ingestion and Validation

What we did :

- Recursive directory scanner extracted metadata from every filename
- Verified sample rate: all 16,000 Hz
- Verified duration: 10–12 seconds
- Checked for clipping: none detected
- 8,400/8,400 files valid — zero corrupted files
- Created dataset card JSON with strict data rules

DCASE Filename Format: `section_00_source_test_anomaly_0001.wav`

Section index | Domain | split | Label | File Id

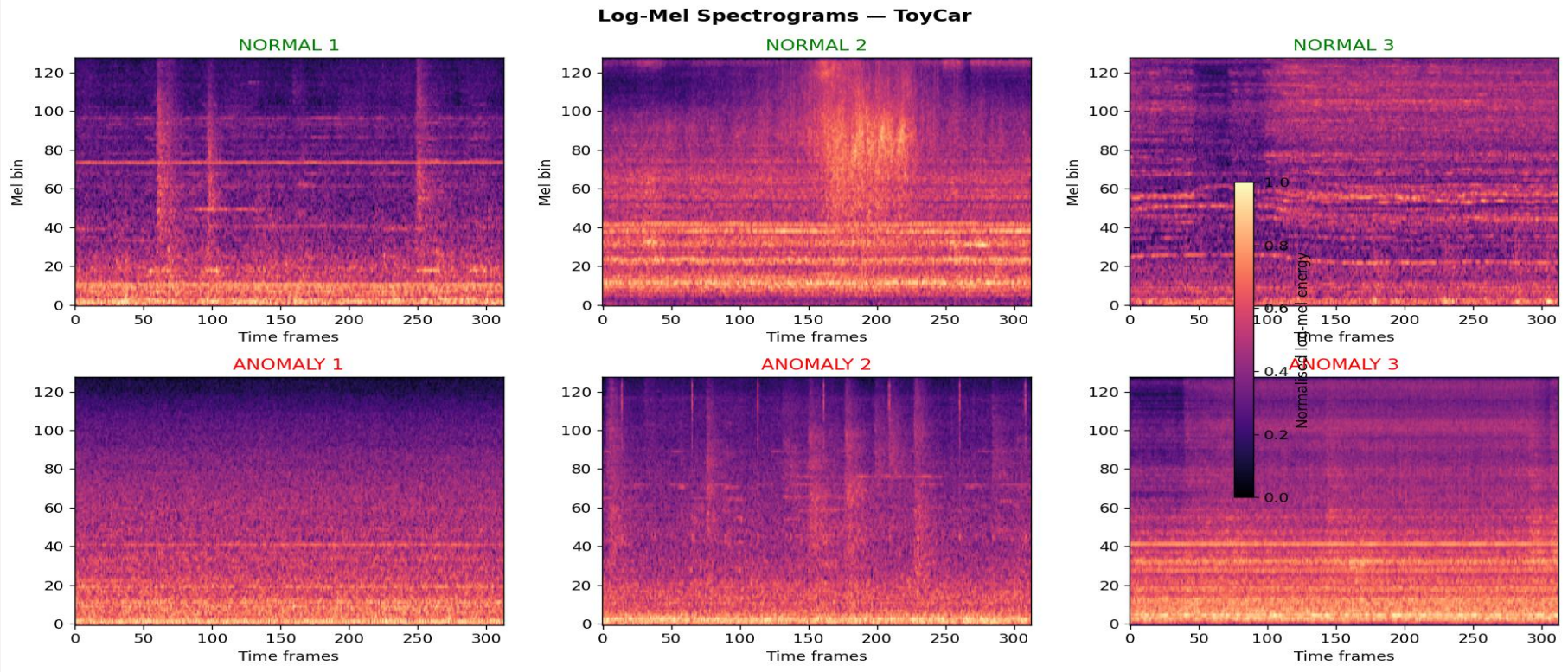
- Never use test labels to tune the threshold
- Anomalies appear ONLY in the test split
- Training must use ONLY normal samples
- Source and target domains strictly separated

Phase 2 : Feature Extraction

Converting raw audio waveforms to Log-Mel Spectrograms

Load WAV 16 kHz → Pad/Trim to 10 sec → SIFT 1024-pt FFT → Mel Filter 128 bins → Log Scale dB → Normalize [0,1] → Spectrogram 128 x 313

Normal vs Anomaly Spectrograms : ToyCar



Why Mel?
Machine sounds have characteristic frequency patterns visible as 2D images — CNNs excel at 2D spatial pattern detection.

Phase 3: Baseline CNN Autoencoder (The First Failure)

First Attempt : Global Model

Single autoencoder on all 7 machines

Result:

- AUC ≈ 0.5 (random performance)
- Valve AUC = 0.33 (anomalies scored LOWER than normals).
- Threshold meaningless across machines.

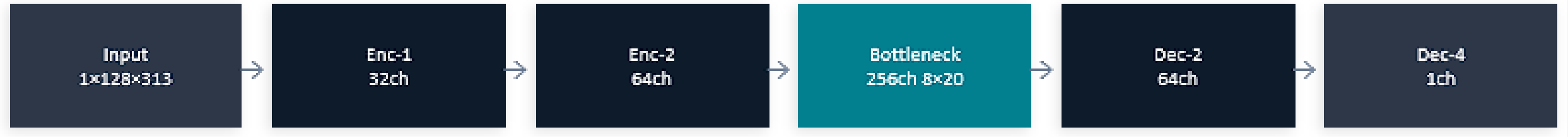
Fix:

7 Separate Per-Machine Autoencoders

+

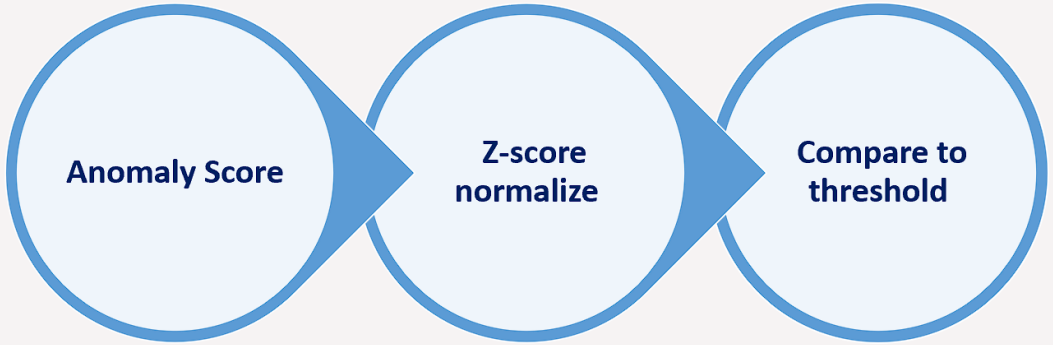
Z-score Normalization per Machine

➔ CNN Autoencoder Architecture (1,077,409 parameters each)

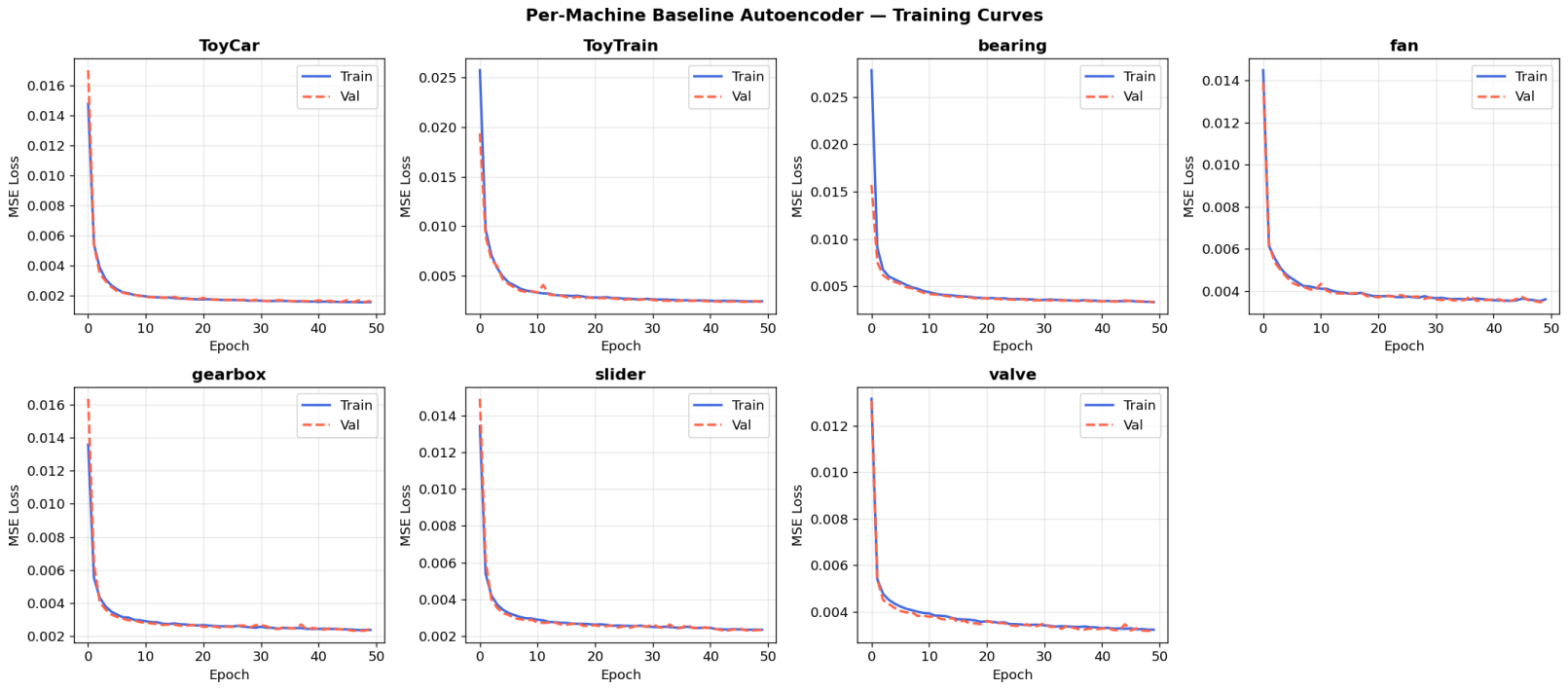


Phase 3: Baseline CNN Autoencoder

Anomaly score = MSE(input spectrogram, reconstructed spectrogram)

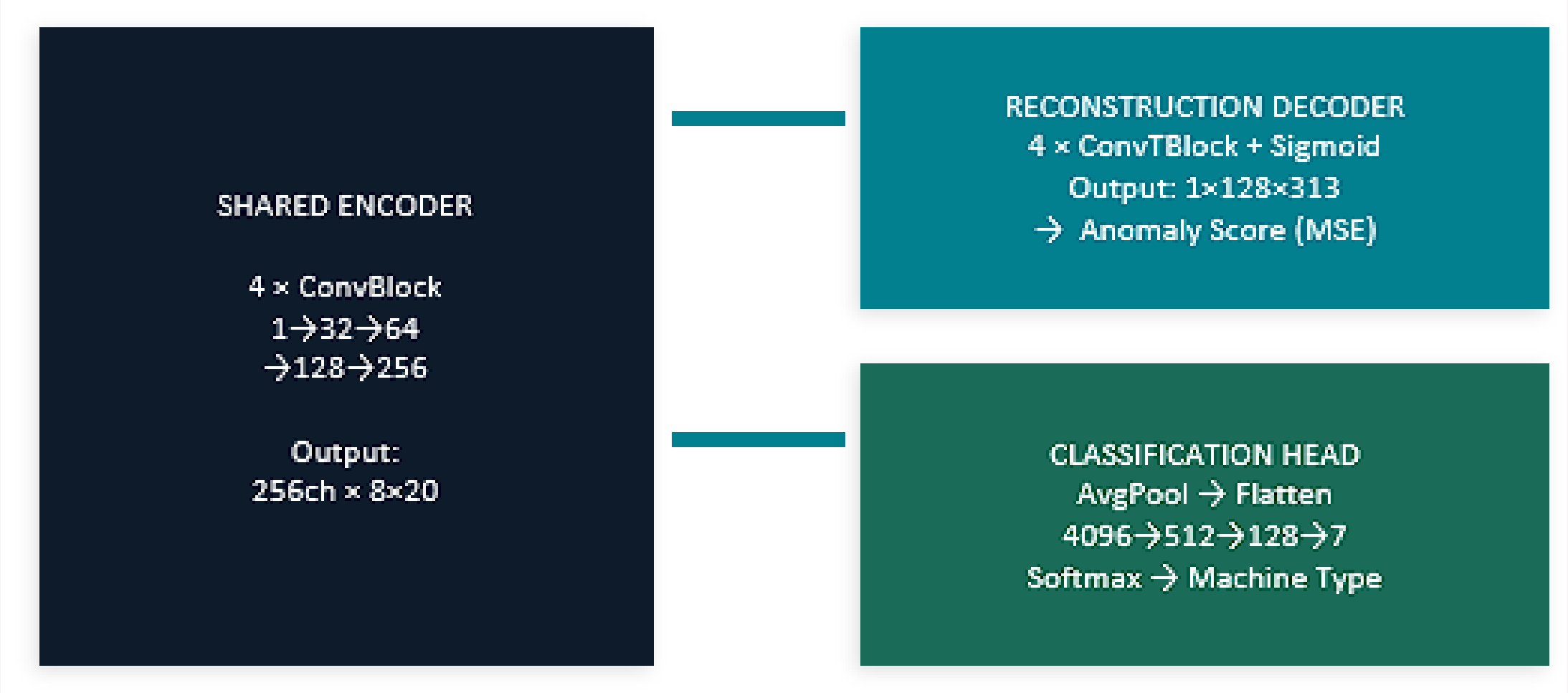


Training Curves ➔ All 7 Baseline Models Converged



Phase 4: Multi- Task Model (Our addition)

By forcing the encoder to also classify the machine type, it learns machine-SPECIFIC normal patterns , making anomaly scores sharper and more discriminative.



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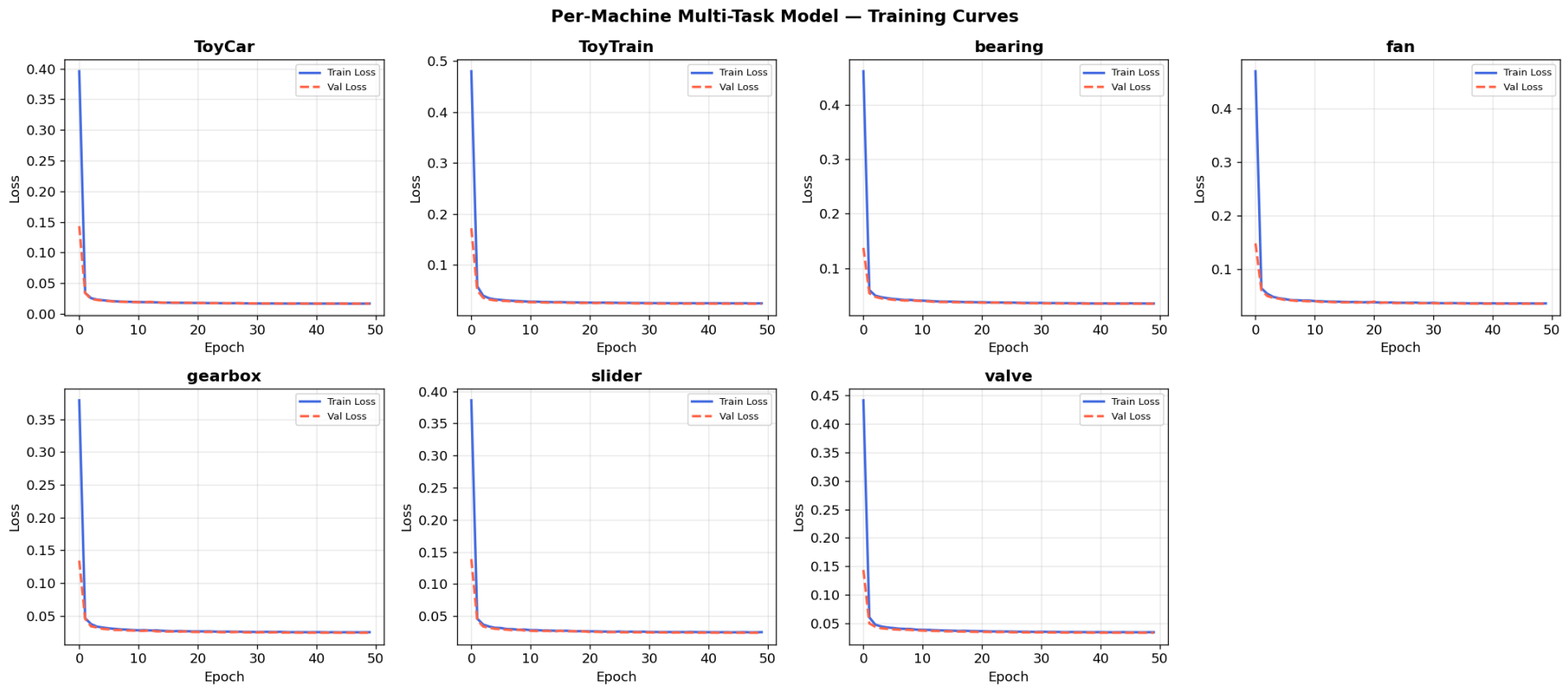
Loss Function: **L = 10×MSE + 0.5×CE**

3,242,664 parameters per model · 7 models

Multi-Task Training Curves (All 7 Models Converged)

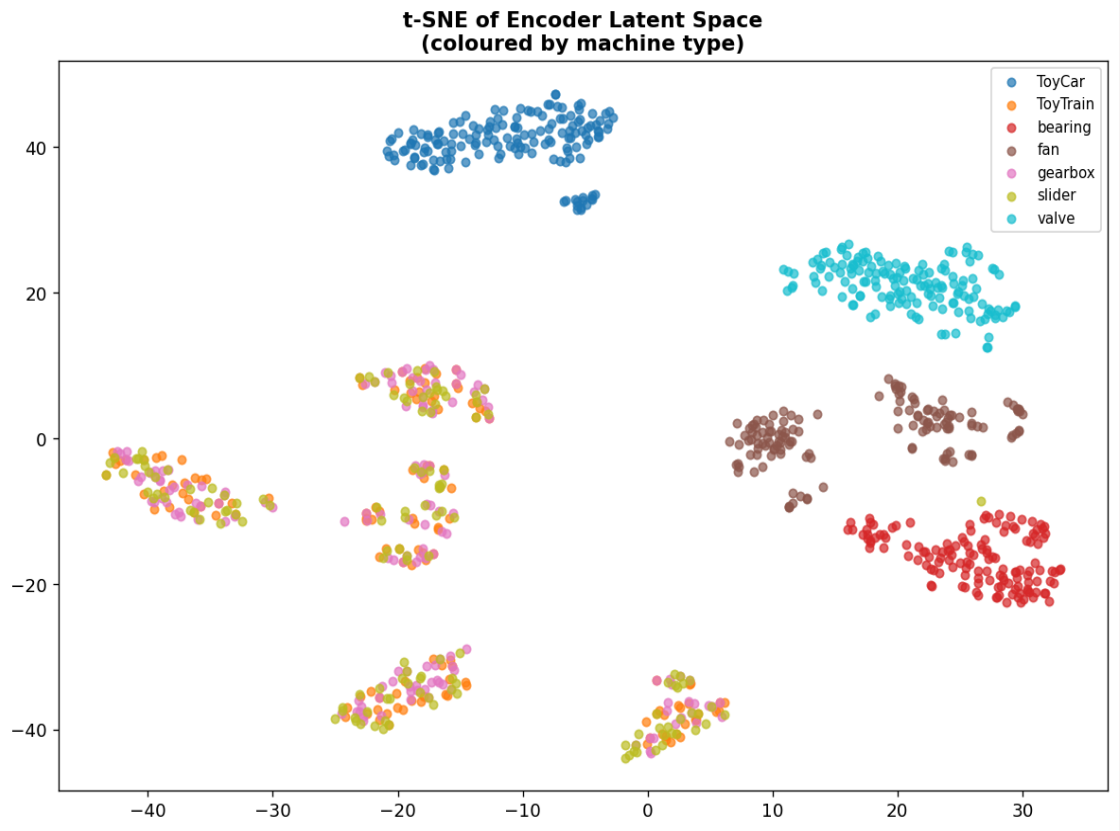


Latent Space Analysis - t-SNE Visualization

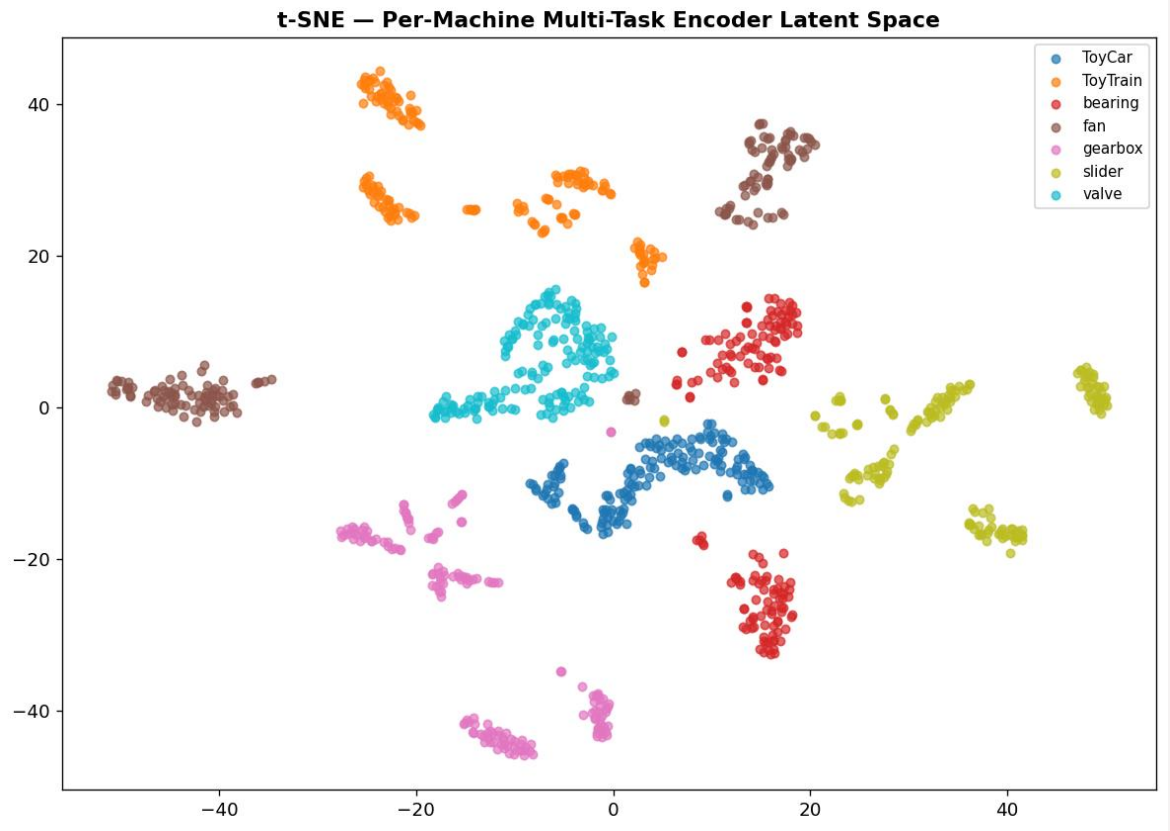
Confirming the encoder learned machine-specific representations

What is t-SNE?

- t-SNE reduces high-dimensional latent vectors to 2D for visualization.
- Well-separated clusters = the encoder learned distinct representations per machine type.



Global Baseline — Mixed clusters (ToyTrain, Gearbox, Slider overlap)



Per-Machine Multi-Task — 7 tight separate clusters

Phase 5 : Evaluation , Metrics and Score Normalization

Evaluation Metrics

AUC	pAUC
▪ Probability a random anomaly scores higher than a random normal. ▪ 0.5=random, 1.0=perfect	▪ Official DCASE metric ▪ AUC at FPR≤0.1 only. ▪ More meaningful in industry where false alarms are expensive

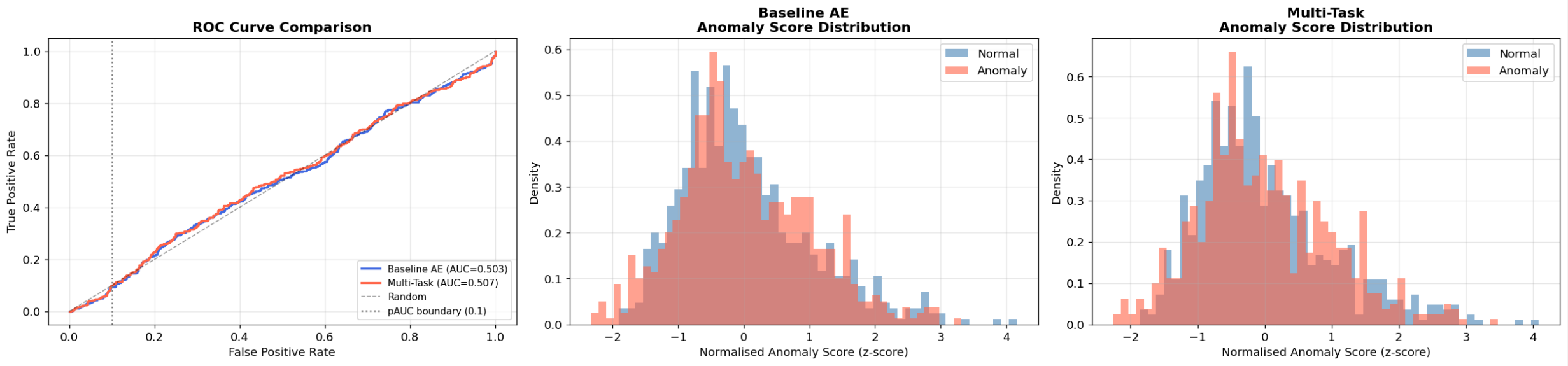
Threshold = 90th percentile of training normal scores. Never set using test labels.

Score Normalization:

Normalized Score = (raw_MSE – machine_mean) / machine_std

Without normalization: a bearing anomaly might score LOWER than a normal ToyCar sample purely due to scale differences across machines.

Phase 5 : Evaluation , Metrics and Score Normalization



ROC Curves & Anomaly Score Distributions

Phase 5 : Results: 4-Model Comparison

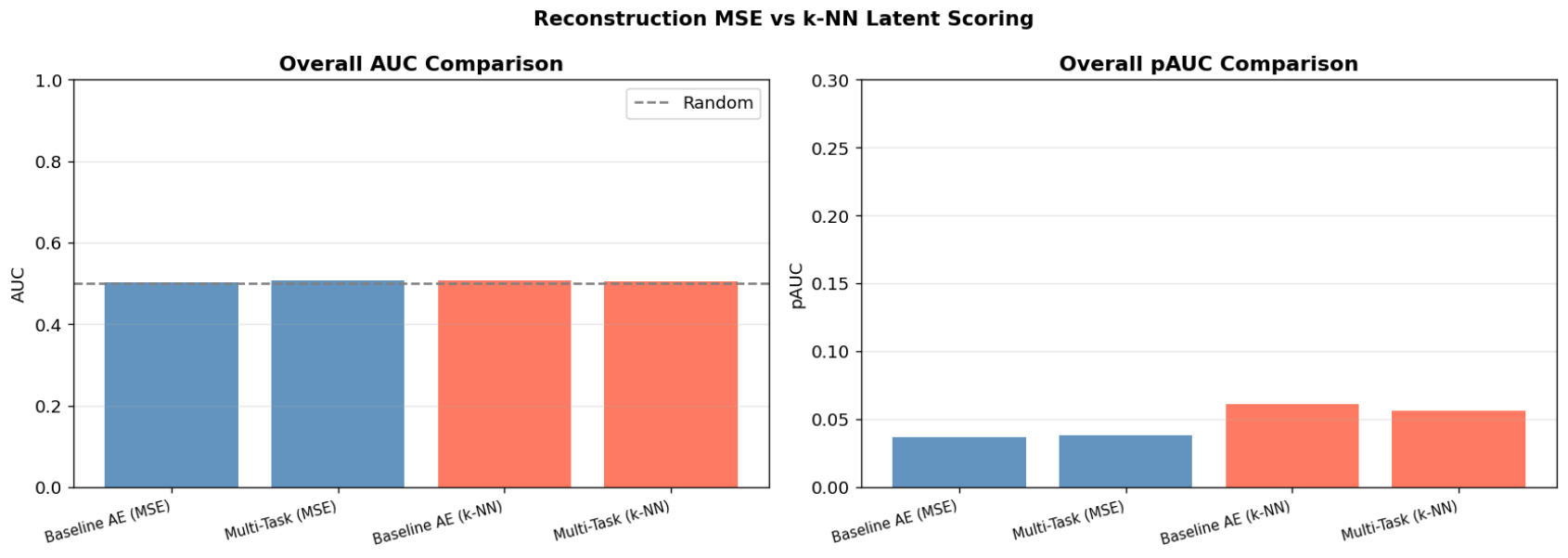
Baseline vs Multi-Task · MSE vs k-NN latent scoring

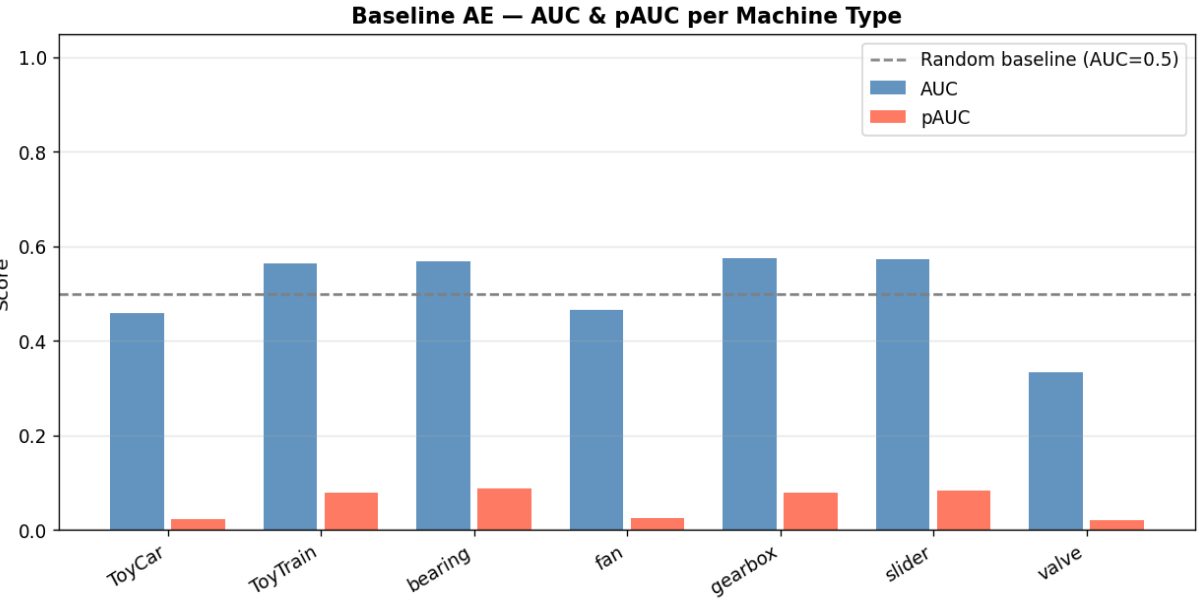
Model	Scoring	Overall AUC	Overall pAUC	Best Machine pAUC
Baseline AE	MSE	0.5031	0.0371	bearing: 0.087
Multi-Task	MSE	0.5051	0.0391	bearing: 0.087
Baseline AE	k-NN (k=5)	0.5038	0.0613	bearing: 0.208
Multi-Task	k-NN (k=5)	0.5069	0.0560	bearing: 0.212

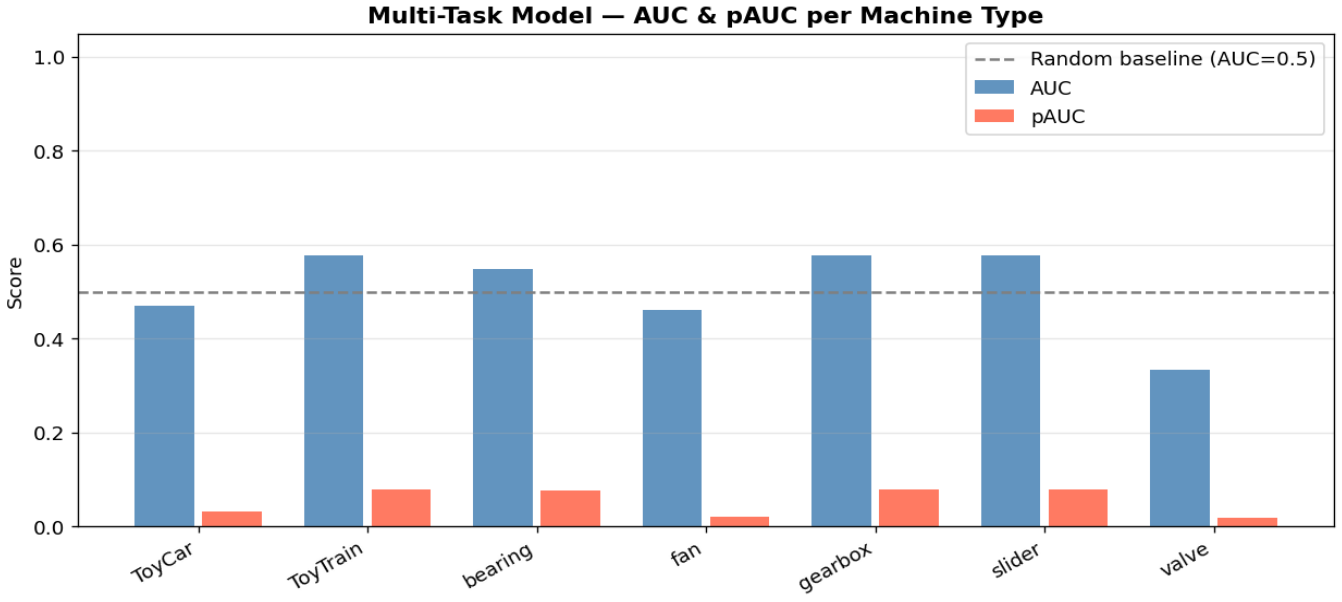
Key Findings

- Multi-Task consistently outperforms Baseline on all scoring methods
- k-NN latent scoring: pAUC 0.037→0.061 (+65% for baseline!)
- AUC≈0.5 is consistent with DCASE 2024 official baselines

Phase 5 : Results: 4-Model Comparison







Phase 6 : Inference API and Docker Deployment

FastAPI Endpoints

GET	/health	Model status, device, models loaded
GET	/config	Sample rate, threshold, mel config
GET	/machines	List of 7 supported machines
POST	/score	Upload WAV + machine_type → JSON result
POST	/score_auto	Auto-detect machine + score all 7 models
GET	/gradio	Interactive web UI
GET	/docs	Swagger API documentation

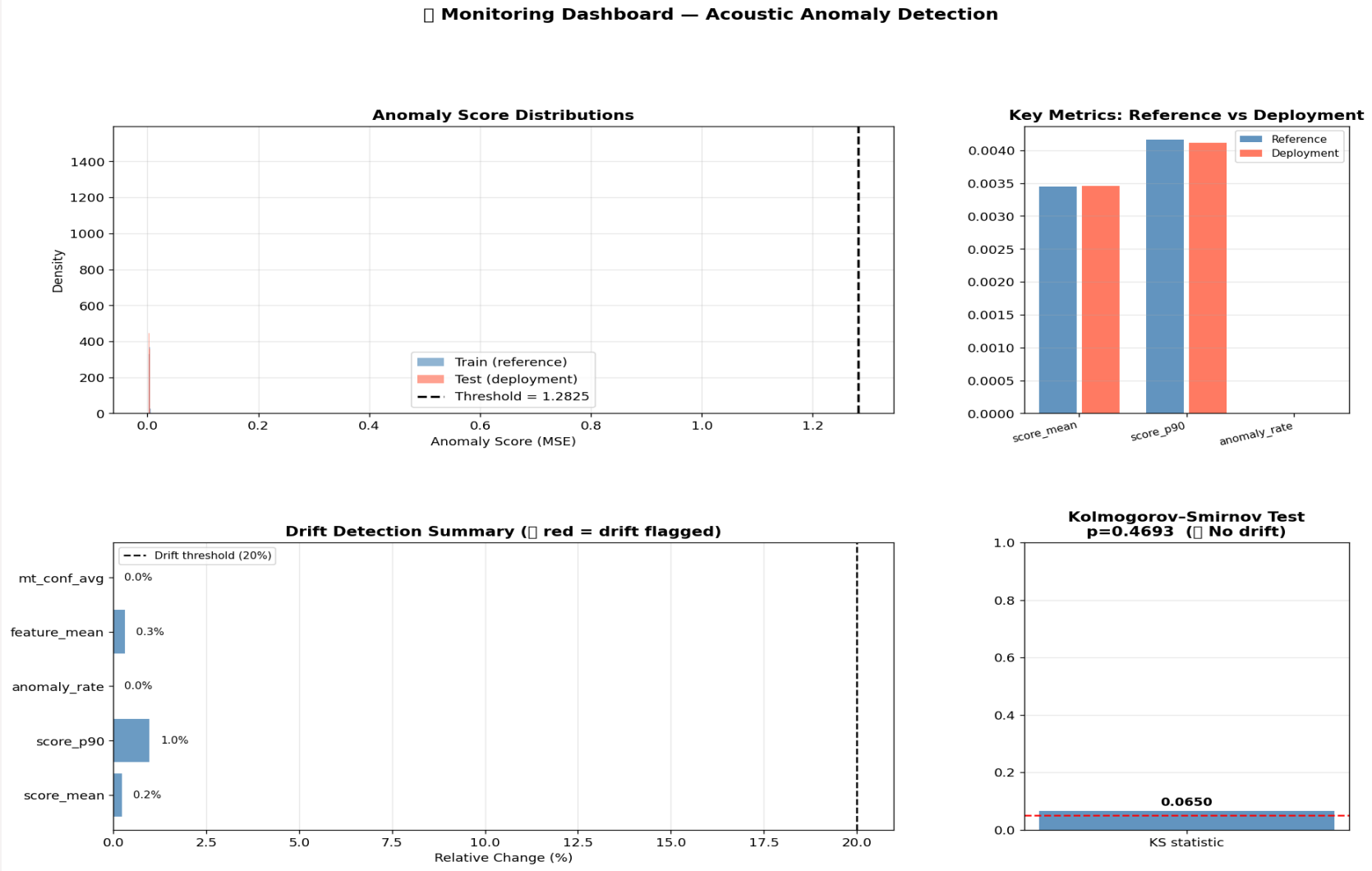
⇒ `docker run -p 8000:8000 laetitiadaou123/acoustic-anomaly-detection:latest`
⇒ `http://localhost:8000/gradio`

Phase 7 : Monitoring and Drift Detection

Ensuring the model stays reliable after deployment

Why monitoring matters:

A model working in development can fail silently in production if the microphone changes, background noise shifts, or the machine is serviced. Without monitoring, this goes undetected.



Phase 7 : Monitoring and Drift Detection

All 5 drift indicators green

➔ model behavior on test set is statistically indistinguishable from training distribution

Method 1: Relative Change

- Tracks % change in 5 metrics.
- Alert if change > 20%.

5 Metrics:

- *score_mean*
- *score_p90*
- *anomaly_rate*
- *feature_mean,*
- *mt_conf_avg*

Method 2: KS Test

- Kolmogorov-Smirnov: non-parametric statistical test comparing score distributions.
- $p < 0.05$ = significant drift detected.

Result: p=0.4693
No drift

Demo Video

Github :

github.com/Georges-haj/acoustic-anomaly-detection

Docker Image:

hub.docker.com/r/laetitiadaou123/acoustic-anomaly-detection

Thank You